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Source: *Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences*, Nov. 21, 1972, Vol. 331, No. 1584 (Nov. 21, 1972), pp. 57-83

Published by: Royal Society

Stable URL: <https://www.jstor.org/stable/78402>

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Foundations of general relativistic high-pressure elasticity theory

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(Communicated by Sir Fred Hoyle, F.R.S. – Received 27 March 1972)

A fully relativistic formulation of the concept of a perfectly elastic solid is developed with a view to application to such problems as (a) the interaction of gravitational radiation with planetary type bodies such as the Earth, and (b) the vibrations and deformations of neutron star crusts. For applications of the former kind a low-pressure Hookean idealization (i.e. a theory postulating a linear stress-strain relation) will often be sufficient. A prototype version of a relativistic Hookean theory suitable for this purpose has already been given by Rayner (1963), but it will be shown that Rayner's theory needs minor corrections in order to be strictly self consistent. For applications to neutron stars a purely Hookean theory will in any case be inadequate and to replace it a high-pressure quasi-Hookean idealization is described. This idealization includes the important special case of the perfect (i.e. isotropic) solid in which the elastic properties may be described in terms of just two (nonlinear) functions of state, namely the pressure and the modulus of rigidity. The theory reduces to the familiar perfect fluid idealization in the limit when the modulus of rigidity is zero.

The discussion is conceptually based on the use of two distinct manifolds: a four-dimensional manifold $\mathcal{M}$ of space-time events, with a fundamental pseudo-Riemannian metric, and a three-dimensional manifold $\mathcal{X}$ of idealized particles, with a fundamental measure representing a conserved particle number density. These manifolds are related by a canonical projection which sends world-lines in $\mathcal{M}$ onto points in $\mathcal{X}$.

1. INTRODUCTION

In recent years a considerable amount of interest and attention has been directed towards the detection of gravitational radiation by its interaction with elastic solid matter, and at the same time a similar current of research has been directed towards the properties of neutron stars including in particular the mechanical behaviour of their solid crusts. As a result, a precise formulation of the principles of elasticity theory in general relativity theory has become overdue.

When it is borne in mind that classical elasticity theory (for which tensor analysis was invented, and in which it finds its most sophisticated applications outside general relativity) is naturally suited to a fully covariant formulation, it is rather surprising that its union with general relativity has been delayed so long. Our purpose here is to remedy this situation by showing that the concept of perfect elasticity can indeed be expressed in a natural and very beautiful way within the framework of general relativity.

A suitable mathematical formalism for the description of a general continuous medium in general relativity has already been developed by Oldroyd. However, the only significant work on the development of a fully relativistic theory of perfect elasticity seems to have been the reformulation of the special relativistic Hergloz

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formalism by De Witt (1962) and the theories of Rayner (1963) and Bennoun (1965). Neither Rayner's theory nor Bennoun's can be judged to be completely successful. Rayner's theory had in any case a limited objective: to develop a *Hookean* perfect elasticity theory (i.e. one with a linear stress-strain relationship). However, although Rayner's theory satisfies Oldroyd's criteria for rheological invariance (which an earlier completely unsound theory suggested by Synge (1959) does not), it will be shown (in § 4) that his attempt to guess a form for the equations of motion consistent with the requirement for perfect elasticity did not quite achieve its purpose. Thus although Rayner's theory is perfectly satisfactory in the small strain limit, it is not strictly consistent with energy conservation when applied to finite strains. Bennoun's work was aimed at the development of a general (not necessarily linear) elasticity theory, and unlike Rayner's it dealt correctly with the integrability conditions imposed by the Bianchi identities. However, as was pointed out by Hernandez (1970), Bennoun was unable to create a fully developed general elasticity theory because he laboured under the serious misconception (which originated in the work of Synge (1959) that it is only possible to discuss *time rates* of strain but not *absolute* strains in general relativity. Thus instead of eliminating the inconsistencies in Rayner's Hookean theory (which, as we shall show, can easily be done) Bennoun could not actually develop any Hookean theory at all. In fact, as was implicit in Rayner's work, and as has been emphasized by Hernandez, the concept of absolute strain does *not* present any serious difficulty in principle in general relativity. Difficulties of the kind Synge and Bennoun appear to have had in mind could only be important if the gravitational forces were so intense as to be dominant on a *microscopic* scale. However, even in the hyperon core of a neutron star, where the gravitational forces must be regarded as highly relativistic on a macroscopic scale, they will remain negligibly small in comparison with local electromagnetic or strong interaction forces between particles, and hence will not directly affect the local crystal (or fluid) properties.

The plan of the present work is as follows. In the second section the kinematics of a general continuous medium is analysed and the concept of convected differentiation (first introduced by Oldroyd (1970)) is explained. In the third section the concept of perfect elasticity (which is often referred to in the classical literature as hyper elasticity) is made precise within the framework of general relativity. In the fourth section this general concept is applied to the construction of an exactly, self-consistent Hookean perfect elasticity theory (i.e. an idealized model theory incorporating a linear stress-strain relation) which agrees with Rayner's prototype theory in the small strain limit. In the fifth section a quasi-Hookean perfect elasticity theory suitable for application to high-pressure neutron star matter is constructed, based on linearization with respect to a shear tensor instead of a strain tensor.

In the sixth section the concept of a perfect solid, i.e. a solid whose intrinsic elastic properties are isotropic, is considered, and the corresponding quasi-Hookean specialization is formulated.

The formalism which we shall describe is general relativistic in the broad sense,

since it does not depend on the precise form of Einstein's equations but depends only on the assumption that the energy-momentum tensor of the material satisfies the standard covariant conservation law. It follows that it is applicable, within the framework of *all* the theories of gravity classified as viable by Thorne, Will & Ni (1971).

2. KINEMATICS OF A CONTINUOUS MATERIAL MEDIUM

Any description of matter in terms of a continuous medium (elastic or otherwise) is based at least implicitly on the use of a 3-dimensional differential manifold, $\mathcal{X}$ say, whose points are to be thought of as the idealized particles of the medium. This manifold will not in general have any canonically defined metric since the physical distances between the particles vary as the particles move, and therefore we shall not have a canonical equivalence between covariant and contravariant vectors on $\mathcal{X}$, i.e. it will not be possible to raise and lower tensor indices in $\mathcal{X}$. However, it is nearly always convenient to assume that $\mathcal{X}$ is at least endowed with a canonical measure, representing the number density of conserved idealized particles. This measure can be represented locally by a totally antisymmetric third-order covariant pseudo-tensor n_{abc} , i.e. a tensor on $\mathcal{X}$ such that

$$dN = n_{abc} dX^a \wedge dX^b \wedge dX^c, \quad (2.1)$$

$$n_{abc} = n_{[abc]}, \quad (2.2)$$

where dN is the number of idealized particles in the volume element

$$dX^1 \wedge dX^2 \wedge dX^3$$

where X^a , $a = 1, 2, 3$ are local coordinates on $\mathcal{X}$. Here, as throughout this work, we use square brackets for antisymmetrization and round brackets for symmetrization of indices. In the application we have in mind, this conserved particle number density is meant to be identified with the baryon number density, but for the purposes of the general theoretical discussion given here the interpretation is not important, since all the important equations including, for example, the definition of the elasticity tensor, are invariant under continuous renormalizations of this measure. The reason for introducing such a measure at the outset rather than in the final stages of application is to avoid the necessity for using tensor densities in addition to ordinary tensors (or equivalently to avoid the frequent appearance of the Jacobean determinants which clutter the fundamental formulae in many standard treatises on elasticity).

In Newtonian theory the manifold $\mathcal{X}$ representing the medium is related to the Galilean space-time manifold by a differentiable one parameter family of imbeddings, parametrized by the Galilean time t , of $\mathcal{X}$ into the 3-dimensional Euclidean space-section at time t . In the relativistic theory to be given here, space-time will be represented by a general 4-dimensional differentiable manifold $\mathcal{M}$ with a general pseudo-Riemannian Lorentz signature metric of the form

$$ds^2 = g_{ab} dx^a dx^b, \quad (2.3)$$

$$g_{ab} = g_{(ab)}. \quad (2.4)$$

Here x^a ($a = 1, 2, 3, 0$) are local coordinates on $\mathcal{M}$ and g_{ab} is a non-singular tensor which, together with its inverse, g^{ab} , defined by

$$g^{ab}g_{bc} = g^a_c \tag{2.5}$$

(where g^a_c is the Kronecker tensor on $\mathcal{M}$), will be used for lowering and raising of tensor indices on $\mathcal{M}$, and for the definition of covariant derivations on $\mathcal{M}$ (which will be denoted by a semi-colon) in the customary manner. Since $\mathcal{M}$ will not in general have a canonical time coordinate t (indeed it need not have any global time

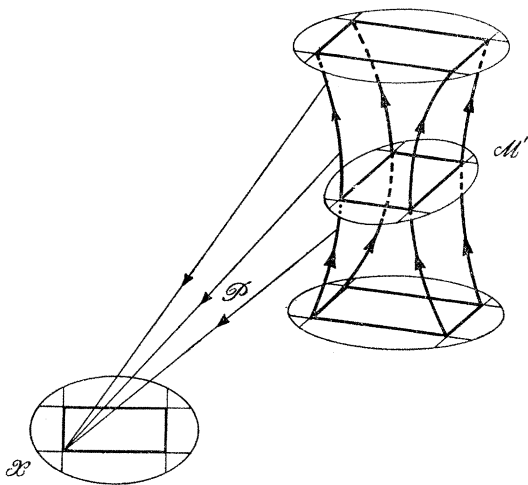


FIGURE 1. The diagram illustrates the projection $\mathcal{P}$ from the 4-dimensional manifold $\mathcal{M}'$ (the part of space-time occupied by the medium) onto the 3-dimensional manifold $\mathcal{X}$ (whose elements represent the idealized particles of the medium). World lines of particles in $\mathcal{M}'$ are represented by heavy lines, with arrows in the direction of the flow vector u^a . Co-moving coordinate surfaces enclosing the co-moving parallelepiped enclosed by the world lines, are marked. Such a parallelepiped represents a convected orthogonal 3-frame in $\mathcal{M}$.

coordinate at all), it is preferable to describe the relationship between $\mathcal{X}$ and $\mathcal{M}$ not by imbeddings of $\mathcal{X}$ into $\mathcal{M}$ as in the Newtonian case, but rather by a canonical differentiable projection (see figure 1)

$$\mathcal{P}: \mathcal{M}' \rightarrow \mathcal{X} \tag{2.6}$$

of an open submanifold $\mathcal{M}' \subset \mathcal{M}$ onto $\mathcal{X}$, where $\mathcal{M}'$ is to be interpreted as the part of space-time through which the medium passes. The projection $\mathcal{P}$ is required to have the properties (1) that the inverse image $\mathcal{P}^{-1}(X) \subset \mathcal{M}$ of any point $X \in \mathcal{X}$ is a timelike line in $\mathcal{M}$ (this line being interpreted as the *world-line* of the idealized particle represented by X), and (2) that any point in any differentially imbedded spacelike hypersurface $\mathcal{H}$ in $\mathcal{M}$ has a neighbourhood $\mathcal{U} \subset \mathcal{H}$ for which the induced mapping $\mathcal{P}: \mathcal{U} \rightarrow \mathcal{X}$ is a differentiable imbedding.

For many applications it is useful to describe the relationship between $\mathcal{X}$ and $\mathcal{M}$ locally by means of co-moving coordinate systems. A coordinate patch

x^a ($a = 1, 2, 3, 0$) on $\mathcal{M}$ is said to be *co-moving* if it is related to some fixed coordinate patch X^a ($a = 1, 2, 3$) on $\mathcal{X}$ by the condition that when $x = x(x^1, x^2, x^3, x^0) \in \mathcal{M}$ then $\mathcal{P}(x) = X(X^1, X^2, X^3) \in \mathcal{X}$ with $X^1 = x^1$, $X^2 = x^2$, $X^3 = x^3$. Such a coordinate patch is said to be *properly* co-moving if $x^0 = \tau$ where τ is a time coordinate measuring proper time along the world lines, i.e. where $ds^2 > 0$ whenever $d\tau = 0$ and where $ds^2 = -d\tau^2$ whenever $dx^1 = dx^2 = dx^3 = 0$.

The projection $\mathcal{P}$ determines a fundamental field of *flow vectors* u^a tangent to the world lines in $\mathcal{M}$ and normalized by the condition

$$u^a u_a = -1, \quad (2.7)$$

and this in turn defines a fundamental field of *orthogonal projection tensors* γ_{ab} defined by

$$\gamma_{ab} = g_{ab} + u_a u_b. \quad (2.8)$$

This tensor acts as a positive metric tensor on the tangent subspace orthogonal to the flow vectors, the corresponding inverse metric tensor being given simply by its equivalent contravariant form, since it satisfies

$$\gamma^{ab} \gamma_{bc} = \gamma^a_c, \quad (2.9)$$

as well as the orthogonality condition

$$\gamma_{ab} u^b = 0. \quad (2.10)$$

Thus for the special case of *orthogonal* space-time tensors, i.e. tensors on $\mathcal{M}$ whose contractions with the flow vector u^a all vanish, γ_{ab} and γ^{ab} can be used in place of the metric tensors g_{ab} and g^{ab} for lowering and raising indices. The orthogonal projection tensor γ_{ab} is the relativistic analogue of the (left) Cauchy–Green tensor which plays a fundamental role in the classical theory of a continuous medium (cf. e.g. Truesdell & Noll 1965). Since it determines the distances between neighbouring particles in their local rest frame, γ_{ab} specifies what will be referred to as the local *state of strain* of the medium.

For conceptual clarity, it is desirable to think of equations of state in terms of what we shall refer to as *material tensor fields*, i.e. tensor fields defined on $\mathcal{X}$, rather than in terms of *space-time tensor fields*, i.e. tensor fields defined on $\mathcal{M}$. The two kinds of fields are related by a fundamental equivalence between the class of all material tensor fields and a certain subclass of space-time tensor fields, which we shall refer to as *materially constant orthogonal space-time tensor fields*. This equivalence is determined by $\mathcal{P}$ as follows.

Clearly the projection $\mathcal{P}$ induces a canonical mapping $\mathcal{P}^{-1}$ of the space of covariant vectors, i.e. differential one-forms, at any fixed point $X \in \mathcal{X}$ into the space of covariant vectors at any fixed point $x \in \mathcal{M}$ such that $\mathcal{P}(x) = X$; moreover, this mapping is an isomorphism onto the subspace of one forms orthogonal to the flow tangent vector u^a at x . Conversely, the projection $\mathcal{P}$ induces a canonical mapping of the space of contravariant vectors, i.e. tangent vectors, at $x \in \mathcal{M}$ onto the corresponding space of tangent vectors at $X = \mathcal{P}(x)$ in $\mathcal{X}$; moreover if this mapping is

restricted to act on the subspace of tangent vectors orthogonal to the flow vector u^a at x , it is one-one, and hence an isomorphism, onto the space of tangent vectors at $\mathcal{P}(x)$. By extending these considerations to general mixed tensors we see that the projection induces a canonical isomorphism

$$\mathcal{P}: T(x) \leftrightarrow \mathcal{P}(T)(X) \quad (2.11)$$

from each vector space of *orthogonal space-time tensors* at any fixed point $x \in \mathcal{M}$ onto the corresponding space of *material tensors* $\mathcal{P}(T)$ at the corresponding fixed point $X = \mathcal{P}(x) \in \mathcal{X}$.

By introducing the concept of materially constant fields we can extend this isomorphism naturally from tensors at fixed points to tensor fields. Thus an orthogonal tensor field T on $\mathcal{M}$ will be said to be *materially constant* if the image $\mathcal{P}(T(x))(X)$, $X = \mathcal{P}(x)$ in $\mathcal{X}$ of its value $T(x)$ at a point $x \in \mathcal{M}$ is independent of the choice of x on the world line of X , i.e. if $\mathcal{P}(T(x))(X)$ is the same for all $x \in \mathcal{P}^{-1}(X)$. It is important to notice that this concept distinguishes between raised and lowered indices: thus if the contravariant form of a certain tensor field is materially constant its covariant form in general will not (and vice versa) except in the special case where the fundamental projection tensor γ_{ab} is itself materially constant, in which case the medium will be said to be in *constant strain state* or equivalently to have a *rigid motion*. (This definition of rigidity is equivalent to that originally given by Born. Rigid motion in general relativity is discussed in detail by Trautman (1964).) It is clear from this definition that the isomorphism (2.11) determines a corresponding *canonical isomorphism from the class of all material tensor fields on $\mathcal{X}$ onto the class of materially constant orthogonal space-time tensors on $\mathcal{M}$.*

We shall take advantage of this isomorphism to avoid introducing a special notation (e.g. by making a distinction between Greek and Latin suffices) to distinguish tensor fields on $\mathcal{X}$ from tensor fields on $\mathcal{M}$. Thus *all* of the tensor field expressions and equations we shall write will be capable of interpretation in terms of 4-dimensional space-time tensor fields on $\mathcal{M}$. *Some* of them, namely those containing only materially constant orthogonal space-time tensor fields, will be able to be (and in most such cases will be intended to be) interpreted also in terms of the corresponding material tensor fields on $\mathcal{X}$.

We can illustrate the use of this isomorphism by applying it to the only material tensor field which has been introduced so far, namely the fundamental particle density measure n_{abc} .

The corresponding materially constant space-time tensor field n_{abc} in $\mathcal{M}$ (obtained by applying $\mathcal{P}^{-1}$ to the material tensor field n_{abc} in $\mathcal{X}$) will of course satisfy

$$n_{abc}u^c = 0. \quad (2.12)$$

The four-dimensional space-time manifold $\mathcal{M}$ will also have a fundamental measure represented by a pseudo-tensor ϵ_{abcd} , determined locally up to a choice of sign by the metric g_{ab} , from the conditions

$$\epsilon_{abcd}\epsilon^{abcd} = 4!, \quad (2.13)$$

$$\epsilon_{abcd} = \epsilon_{[abcd]}. \quad (2.14)$$

This in turn will determine a corresponding fundamental space-volume density measure ϵ_{abc} on the three-dimensional tangent subspaces orthogonal to the flow lines by the condition

$$\epsilon_{abc} = \epsilon_{abcd} u^d, \quad (2.15)$$

which automatically implies that the antisymmetry condition

$$\epsilon_{abc} = \epsilon_{[abc]} \quad (2.16)$$

and the orthogonality conditions

$$\epsilon_{abc} u^c = 0$$

are satisfied. By contracting this volume density measure with the particle number density measure we obtain a scalar function n on $\mathcal{M}$ given by

$$n = \epsilon^{abc} n_{abc}, \quad (2.17)$$

which represents the number density of conserved idealized particles of the medium per unit metric volume. The requirement that the sign of n be positive determines the sign of ϵ_{abcd} locally in terms of that of n_{abc} . (These signs can of course only be assigned globally if appropriate orientability requirements are satisfied.) In most applications n can conveniently be taken to be simply the baryon density.

Just as a materially constant orthogonal space-time tensor field on $\mathcal{M}$ can be identified with a *fixed* material tensor field on $\mathcal{X}$, so also a general orthogonal space-time tensor field T on $\mathcal{M}$ can be identified, by means of (2.11), with a corresponding material tensor field *function of time* on $\mathcal{X}$, by a correspondence of the form $T(x) \leftrightarrow \mathcal{P}(T)(X, \tau)$, where $X = \mathcal{P}(x)$ and $\tau = \tau(x)$ is a proper time coordinate function on $\mathcal{M}$, i.e. a scalar field on $\mathcal{M}$ whose gradient is timelike and such that $\tau_{,a} u^a = 1$. Such a correspondence is not of course canonical since it depends on the choice of τ . However, it leads to a canonical definition of the *convected differential* $dT(x)$ of an orthogonal space-time tensor field T on $\mathcal{M}$ at a particular point $x \in \mathcal{M}$, corresponding to transport by a distance $d\tau$ along the world line through x , according to the formula

$$\mathcal{P}(dT(x)) = d\mathcal{P}(T)(X, \tau), \quad (2.18)$$

where $d\mathcal{P}(T)(X, \tau)$ is the corresponding *material differential*, i.e. the differential of the variable material tensor field $\mathcal{P}(T)(X, \tau)$ at the fixed point $X = \mathcal{P}(x)$ in $\mathcal{X}$ due to the variation $d\tau$ in τ evaluated at $\tau = \tau(x)$. This in turn leads to a canonical definition of an orthogonal tensor field on $\mathcal{M}$ which we shall refer to as the *convected derivative* of T , and denote by $[T]'$ according to the definition

$$[T]'(x) = dT(x)/d\tau. \quad (2.19)$$

Clearly the convective derivative of T is zero if and only if T is materially constant.

This concept of the convective derivative was first introduced explicitly by Oldroyd (1970), although its use was implicit in the earlier work of Rayner (1963). The treatment we have given so far (and that of Oldroyd also) applies only to orthogonal space-time tensor fields. However, since a general space-time tensor field T can be represented by an *orthogonal decomposition* as a sum of tensor products of

orthogonal tensor fields with the covariant and contravariant flow vectors u_a and u^a the concepts of convective derivative and material constancy can be extended naturally to T by defining them in terms of the convective derivatives and material constancy of the orthogonal tensors in the orthogonal decomposition of T . Equivalently, the convective derivative of a general space-time tensor field T may be defined by specifying the components of $[T]'$ with respect to a *convected 4-frame* field in $\mathcal{M}$ as the proper time derivatives, along the world lines, of the corresponding 4-frame components of T . By a *convected 4-frame* we mean a 4-frame on $\mathcal{M}$ which is related to a fixed 3-frame on $\mathcal{X}$ as follows: **three of the tangent vectors and also three of the dual covectors of the 4-frame in $\mathcal{M}$ are materially constant orthogonal vector fields, equivalent under $\mathcal{P}$ to the three vectors and the three dual covectors of the 3-frame on $\mathcal{X}$, and the remaining tangent vector and covector of the 4-frame are simply u^a and u_a .**

With this definition, the convected derivative $[T_{a...}]'$ of a general mixed **space-time tensor** $T_{a...}^{b...}$ is given in terms of covariant derivatives by

$$[T_{a...}]' = \dot{T}_{a...}^{b...} - T_{a...}^{c...} (u^b_{;c} + u^b \dot{u}_c) - \dots + T_{c...}^{b...} (u^c_{;a} + u^c \dot{u}_a) + \dots, \quad (2.20)$$

where the negative summation is taken over the contravariant indices of T and the positive summation is taken over the covariant indices of T . We have used the standard convention of placing a dot directly over the tensor symbol to denote covariant differentiation along the flow lines, i.e.

$$\dot{T}_{a...}^{b...} = T_{a...;c}^{b...} u^c. \quad (2.21)$$

This applies in particular to the *acceleration vector* $\dot{u}^a$ defined by

$$\dot{u}^a = u^a_{;b} u^b. \quad (2.22)$$

The square brackets denoting convected differentiation are placed outside the tensor indices to emphasize that (except in the special case of rigid motion) convective differentiation does not commute with raising and lowering of indices. The derivation of the formula (2.20) is given in the appendix.

We remark that when T is an *orthogonal* tensor, the terms containing the acceleration $\dot{u}_a$ in the *positive* summation drop out and (2.20) reduces to Oldroyd's formula. When T is a *covariant orthogonal* tensor the terms containing the acceleration drop out altogether and the convected derivative reduces simply to the *Lie derivative* with respect to the flow vector field u^a . Finally, in the case of a *scalar* the convective derivative and the covariant derivative both reduce simply to the proper-time derivative along the flow lines.

We conclude this section by giving formulae for the convective derivatives of the covariant and contravariant terms of the fundamental orthogonal tensors γ_{ab} and ϵ_{abc} in terms of the fundamental decomposition of the covariant derivatives of the flow field (due to Ehlers 1961) according to which orthogonal tensors ω_{ab} and θ_{ab} (the *rotation tensor* and the *strain rate tensor*) are defined by

$$u_{a;b} = \theta_{ab} + \omega_{ab} - \dot{u}_a u_b, \quad (2.23)$$

$$\text{subject to} \quad \theta_{ab} = \theta_{(ab)}, \quad (2.24)$$

$$\omega_{ab} = \omega_{[ab]}, \quad (2.25)$$

while a further (trace-free) orthogonal tensor σ_{ab} (the *shearing rate* tensor) and a scalar θ (the *expansion rate*) are defined by

$$\theta_{ab} = \sigma_{ab} + \frac{1}{3}\theta\gamma_{ab}, \quad (2.26)$$

$$\text{with} \quad \theta = \theta_a{}^a = \gamma^{ab}\theta_{ab} = u_a{}^{:a}. \quad (2.27)$$

In terms of these quantities we can write

$$[\gamma_{ab}]^{\cdot} = 2\theta_{ab}, \quad (2.28)$$

$$[\gamma^{ab}]^{\cdot} = -2\theta^{ab}, \quad (2.29)$$

$$[\epsilon_{abc}]^{\cdot} = \theta\epsilon_{abc}, \quad (2.30)$$

$$[\epsilon^{abc}]^{\cdot} = -\theta\epsilon^{abc}. \quad (2.31)$$

Using the last of these equations together with the conservation law

$$[n_{abc}]^{\cdot} = 0, \quad (2.32)$$

which follows from the definition of n_{abc} as a materially constant field on $\mathcal{M}$ we obtain (from 2.17) the equation

$$\dot{n} = -\theta n, \quad (2.33)$$

which is equivalent to the familiar expression

$$(nu^a)_{;a} = 0 \quad (2.34)$$

of the local particle conservation law.

The elasticity theory which follows is formulated in terms of various orthogonal space-time tensor fields (of the general form $T_{a...}^{b...}$) which are *functions of strain* in the sense that the components of the corresponding material tensor $\mathcal{P}(T_{a...}^{b...})$ in X are functions of the (six independent) components of $\mathcal{P}(\gamma_{cd})$, in $\mathcal{X}$ (the actual functional form depending of course on the choice of coordinates in $\mathcal{X}$). In other words $T_{a...}^{b...}$ is a function of strain if, on each world line, its components in terms of a convected tetrad are well defined functions of the components of γ_{cd} in this tetrad, the form of these functions depending on the choice of the convected tetrad.)

Clearly if $T_{a...}^{b...}$ is an orthogonal space-time tensor function of strain it will satisfy an equation of motion of the form

$$[T_{a...}^{b...}]^{\cdot} = 2 \frac{\partial(T_{a...}^{b...})}{\partial(\gamma_{cd})} \theta_{cd}, \quad (2.35)$$

where $\partial(T_{a...}^{b...})/\partial(\gamma_{cd})$ is itself an orthogonal space-time tensor function of state, symmetric under interchange of the indices c and d .

3. THE CONCEPT OF PERFECT ELASTICITY AND THE FUNDAMENTAL CONSISTENCY CONDITION

The concept of perfect elasticity (i.e. what is often referred to technically as hyper elasticity) owes its origin in classical theory to Green. It can conveniently and naturally be formulated in relativistic terminology as follows. A material medium will be said to behave *perfectly elastically* if it gives a symmetric contribution T^{ab} to the total energy momentum tensor in the space-time which satisfied the following three conditions:

(1) There is no material energy transport relative to the local rest frames of the medium, i.e. the flow vector u^a is an eigenvector of T^{ab} .

(2) The force density f^a due to other fields (such as an electromagnetic field) which may be present not only obeys the standard energy momentum conservation law

$$T^{ab}{}_{;b} = f^a \quad (3.1)$$

but also satisfies the orthogonality condition

$$f^a u_a = 0. \quad (3.2)$$

(In other words the external fields do no work on the medium.)

(3) For any given material point $x \in \mathcal{X}$ (i.e. along any given particle world-line of the medium) quantities ρ and p^{ab} are *well-defined functions of strain*, in the sense defined at the end of the previous section. This condition carries the implication that the evolution of the medium is thermodynamically reversible (i.e. entropy conserving). Thus effects such as viscosity, hysteresis, etc., are excluded from consideration.

These conditions will very often be applicable when the medium represents a solid, liquid or gas whose motion can be regarded as adiabatic and frictionless. They could be expected to fail, however, if the medium was subject to deformations (e.g. due to ultra-high frequency sound waves) whose characteristic lengthscales were comparable with the microscopic lengthscales determining the crystal or fluid structure of the medium; in such cases T^{ab} could be expected to depend not simply on the local value of the state of strain, defined by γ_{ab} , but also on the spatial strain gradients. Similarly, if the material were subjected to gravitational fields or accelerations so high that the inertial forces on the solid or fluid particles became comparable with the microscopic forces determining their crystal or fluid properties, then it could be expected that an explicit dependence of T^{ab} on the acceleration $\dot{u}^a$ would need to be taken into account. Nevertheless, complications of this sort would arise only in extreme situations where the other assumptions, such as that of adiabaticity, would be on the verge of breaking down anyway (as in the propagation of shock waves). Steady gravitational fields, even in the deep interior of a neutron star, will normally be far too weak to introduce effects of this kind.

A less extreme kind of adiabatic frictionless situation in which these conditions would fail is that where non-geometric internal degrees of freedom, of the medium, corresponding for example to electric or magnetic polarizability, need to be taken

into account. The conditions (1) to (3) are compatible with the presence of a force density of the form $f^a = F^{ab}j_b$ due to an electromagnetic field F^{ab} satisfying the usual equations $F_{(ab)} = 0$, $F_{[ab];c} = 0$, $F^{ab}{}_{;b} = 4\pi j^a$ provided that the medium is *unpolarizable* and *either* an uncharged perfect insulator (which requires that the current density j^a be zero) *or* a perfect conductor (which requires that $F^{ab}u_a$ be zero). However, in a *polarizable* medium even an adiabatic frictionless motion would not necessarily satisfy condition (2), and to deal with such cases a generalization of the present treatment would be necessary.

It follows immediately from condition (1) that T^{ab} can be expressed in terms of a scalar ρ (the *energy density*) and a tensor p^{ab} (the *pressure tensor*, or in other words the *negative of the stress tensor*) in the form

$$T^{ab} = \rho u^a u^b + p^{ab}, \quad (3.3)$$

subject to

$$p^{ab} = p^{(ab)}, \quad (3.4)$$

$$p^{ab}u_b = 0. \quad (3.5)$$

The time development of the flow system in the background manifold $\mathcal{M}$ is determined by the three independent equations

$$(\rho\gamma^{ab} + p^{ab})\dot{u}_b = f^a - \gamma^a{}_c \gamma^b{}_d p^{dc}{}_{;b} \quad (3.6)$$

(obtained from the conservation equation (3.1) by projection orthogonal to the flow lines) which give the acceleration as a homogeneous function of the force density and the space-gradients of the pressure tensor. These three equations for the variation of the three independent components of the flow vector form a complete system when taken in conjunction with the equations of state postulated by condition (3) which specify ρ and p^{ab} in terms of γ_{ab} and thus indirectly, by (2.8), in terms of u^a . (The time development of the metric tensor g_{ab} itself is of course assumed to be given either by the Einstein equations or some alternative system depending on the precise theory of gravity which is being used.) However, the equations of motion (3.1) contain a fourth independent equation (obtained by contraction with the flow vector) which in consequence of equation (3.2) reduces to the form

$$\dot{\rho} = -\rho\theta - p^{ab}\theta_{ab}. \quad (3.7)$$

It is clear that the presence of this superfluous fourth equation of motion will cause the system of equations of motion to be overdetermined unless the equations of state specifying ρ and p^{ab} can be chosen in such a way that it reduces to an identity. It is easy to satisfy this condition however since (3.7) can be expressed equivalently using (2.27) and (2.28) in terms of convected differentials along the flow lines, in the form

$$d\rho + \frac{1}{2}(\rho\gamma^{ab} + p^{ab})d(\gamma_{ab}) = 0. \quad (3.8)$$

This equation can be interpreted alternatively in terms of ordinary differentials of ρ and γ_{ab} on $\mathcal{X}$ regarding ρ and p^{ab} as variable material fields on $\mathcal{X}$ determined by the equations of state as functions of a variable material field γ_{ab} on $\mathcal{X}$. As such (3.8) represents the fundamental *consistency condition* on the equation of state which is clearly necessary and sufficient for (3.7) to be satisfied identically, and hence for the

equations of motion to be well determined. This condition implies that there can only be one independent equation of state (rather than seven) for the ρ and the six independent components of p^{ab} . When ρ has been specified by an equation of state of the form $\rho = \rho(\gamma_{ab})$ in terms of the six independent components of γ_{ab} the corresponding equations of state for the six independent components of p^{ab} as functions of γ_{ab} will automatically be determined by (3.8), in the form

$$p^{ab} = -2 \frac{\partial(\rho)}{\partial(\gamma_{ab})} - \rho \gamma^{ab}, \quad (3.9)$$

where $\partial(\rho)/\partial(\gamma_{ab})$ is a variable contravariant function of γ_{ab} on $\mathcal{X}$ to which of course there will correspond a fixed orthogonal space-time tensor field on $\mathcal{M}$. Since the tensor $\partial(\rho)/\partial(\gamma_{ab})$ is automatically symmetric under interchange of the indices a, b in consequence of the corresponding symmetry of γ_{ab} it follows that the tensor p^{ab} defined by (3.7) is also automatically symmetric as required by (3.4). Thus *any* well-behaved function ρ of γ_{ab} determines corresponding well-behaved equations of state for p^{ab} in terms of γ_{ab} .

For many purposes it is convenient to express the equations in terms of an average mass per particle (or density per baryon) function m defined by

$$\rho = nm \quad (3.10)$$

in place of the density function. If we specify the equations of state by giving m rather than ρ as a function of γ_{ab} we obtain in place of (3.9) the simpler formal expression

$$p^{ab} = -2n \partial(m)/\partial(\gamma_{ab}). \quad (3.11)$$

This leads us to introduce a new bisymmetric tensor function of state E^{abcd} on $\mathcal{X}$, which we shall refer to as the *elasticity tensor*, by the defining relation

$$E^{abcd} = -2 \frac{\partial(p^{ab})}{\partial(\gamma_{cd})} - p^{ab} \gamma^{cd}, \quad (3.12)$$

or equivalently

$$E^{abcd} = -2n \frac{\partial}{\partial(\gamma_{cd})} \left(\frac{p^{ab}}{n} \right). \quad (3.13)$$

This tensor is automatically bisymmetric in the sense that

$$E^{abcd} = E^{(ab)(cd)}, \quad (3.14)$$

the symmetry on the first pair of indices being a consequence of the symmetry of p^{ab} , and the symmetry on the second pair of indices being a consequence of the symmetry of γ_{ab} . Moreover, it follows from the expression (3.11) of the consistency condition that the elasticity tensor must satisfy

$$E^{abcd} = 4n \frac{\partial^2(m)}{\partial(\gamma_{ab}) \partial(\gamma_{cd})}, \quad (3.15)$$

from which it is immediately clear that it must be symmetric under interchange of the first and last pairs of indices, i.e.

$$E^{abcd} = E^{cdab}. \quad (3.16)$$

If the equations of state are specified by giving p^{ab} (rather than ρ or m) in terms of γ_{ab} then the symmetry condition (3.16) clearly represents the *fundamental integrability condition* which is necessary and sufficient for the existence of a function of state m , and hence of the corresponding function ρ which will satisfy the basic consistency condition (3.8). (The tensor E^{abcd} introduced here is the precise analogue of the elasticity tensor traditionally denoted by C^{abcd} in the classical theory of elasticity, in accordance with the precise definition as given, for example, by Truesdell & Noll (1965), page 131, equation (45.2). We have avoided using the traditional symbol C^{abcd} (which was historically suggested by the requirement that elasticity components be constants in the Hookean special case) because in relativity the symbol C^{abcd} is customarily reserved for Weyl's conformal tensor, so that confusion might have arisen in problems concerning the interaction of elastic and gravitational waves.) In consequence of (3.16) the tensor E^{abcd} can only have 21 independent components instead of the 36 which would have been permitted by the immediate symmetries (3.14).

When interpreted as a fixed tensor field on $\mathcal{M}$ rather than as a variable tensor field function of state on $\mathcal{X}$ the elasticity tensor will of course satisfy the orthogonality condition which, in view of the symmetries (3.14) and (3.16) can be expressed by the single equation

$$E^{abcd}u_{;d} = 0. \quad (3.17)$$

It follows directly from the definition (3.12) that the convected derivative of the pressure tensor satisfies

$$[p^{ab}]^* = -p^{ab}\theta - E^{abcd}\theta_{cd}, \quad (3.18)$$

which is analogous to the equation (3.7) satisfied by the derivative of the density.

It is evident that the equations of state may be specified by giving E^{abcd} in terms of γ_{ab} subject to further integrability conditions. When this approach is used the equations (3.7) and (3.18) must be treated as non-trivial equations of motion supplementing (3.6), instead of being merely identities as they are when the equation of state is specified by giving ρ in terms of γ_{ab} .

For many purposes it is convenient to decompose p^{ab} and E^{abcd} into reduced parts as follows. We introduce the *pressure* p and the *shear-stress tensor* q^{ab} by

$$p^{ab} = p\gamma^{ab} - q^{ab}, \quad (3.19)$$

$$\text{with} \quad p = \frac{1}{3}p_a^a, \quad (3.20)$$

$$\text{so that} \quad q_a^a = 0. \quad (3.21)$$

Similarly, we introduce a symmetric tensor D^{ab} and a scalar E given by

$$E^{ab}{}_c = 3D^{ab} + \frac{1}{3}E\gamma^{ab}, \quad (3.22)$$

$$\text{with} \quad E = E_a^a{}_b{}^b. \quad (3.23)$$

This enables us to define a *shear elasticity* tensor S^{abcd} by the decomposition

$$E^{abcd} = S^{abcd} + D^{ab}\gamma^{cd} + D^{cd}\gamma^{ab} + \frac{1}{9}E\gamma^{ab}\gamma^{cd}. \quad (3.24)$$

This definition ensures that in addition to having the same symmetries as E^{abcd} , i.e.

$$S^{abcd} = S^{(ab)(cd)} = S^{cdab}, \quad (3.25)$$

the tensor S^{abcd} will be trace free on the first (and hence also by (3.25) on the last) two indices, i.e.

$$S^{ab}{}^c{}_c = 0. \quad (3.26)$$

The tensor D^{ab} is also trace-free, i.e.

$$D_a{}^a = 0. \quad (3.27)$$

Clearly S^{abcd} has only 15 independent components, and D^{ab} only 5.

In terms of these decompositions we can write the primary equations of motion (3.6) and the equations (3.7) and (3.18) for the time development of ρ and p^{ab} in the form

$$(\rho + p)\dot{u}^a - q^{ab}\dot{u}_b = f^a - \gamma^{ab}p_{,b} + \gamma^a{}_c \gamma^b{}_d q^{cd}{}_{;b}, \quad (3.28)$$

$$\dot{\rho} = -(\rho + p)\theta + q^{ab}\sigma_{ab}, \quad (3.29)$$

$$\dot{p} = -\frac{1}{3}(p + \frac{1}{3}E)\theta - (D^{ab} + \frac{2}{3}q^{ab})\sigma_{ab}, \quad (3.30)$$

$$[q^{ab}]^* = -(q^{ab} - D^{ab})\theta + (S^{abcd} - \frac{2}{3}\gamma^{ab}q^{cd})\sigma_{cd} - 2p\sigma^{ab}. \quad (3.31)$$

These four equations form a complete system of equations of motion when the equations of state are specified by giving S^{abcd} , D^{ab} and E as functions of γ_{ab} . We remark that (3.24) does not represent a complete reduction of E^{abcd} since S^{abcd} is further reducible. We can obtain a complete reduction of S^{abcd} by introducing a symmetric tensor H^{ab} and a scalar S defined by

$$3S^a{}_{c}{}^{bc} = 7H^{ab} + S\gamma^{ab}, \quad (3.32)$$

with

$$S = S^a{}_{ba}{}^b. \quad (3.33)$$

This enables us to define a tensor W^{abcd} by the decomposition

$$S^{abcd} = W^{abcd} + 2H^{a(c}\gamma^{b)d)} + 2H^{b(c}\gamma^{d)a)} - \frac{4}{3}(H^{ab}\gamma^{cd} + H^{cd}\gamma^{ab}) + \frac{1}{5}S(\gamma^{a(c}\gamma^{d)b)} - \frac{1}{3}\gamma^{ab}\gamma^{cd}). \quad (3.34)$$

The tensor W^{abcd} is irreducible. It not only has the same symmetries as S^{abcd} but is in fact totally symmetric, i.e.

$$W^{abcd} = W^{(abcd)}. \quad (3.35)$$

It has no cross trace, i.e.

$$W^a{}_{c}{}^{bc} = 0 \quad (3.36)$$

and thus, by the symmetry (3.35), no traces between the first or last pairs of indices either. The tensor H^{ab} is also trace free, i.e.

$$H_a{}^a = 0 \quad (3.37)$$

so that the expansion (3.34) represents a complete reduction of S^{abcd} . The tensor W^{abcd} has only nine independent components and the tensor H^{ab} only five. These together with the 5-component tensor D^{ab} and the two scalars E and S make up the irreducible parts of the 21-component tensor E^{abcd} .

A particularly simple situation arises when the material is in a state which is

locally *cubically symmetric* at some point in $\mathcal{M}$. At such a point H^{ab} and D^{ab} must clearly vanish. Moreover, W^{abcd} will only be able to have four independent components of which three determine the frame of spherical symmetry, so that it will only have one independent invariant which may be taken to be $W_{abcd}W^{abcd}$. There will be only two other independent invariants of the elasticity tensor namely E and S . The elasticity tensor may be expressed in the form

$$E^{abcd} = \frac{1}{9}E\gamma^{ab}\gamma^{cd} + \frac{1}{3}S(\gamma^{a(c}\gamma^{d)b} - \frac{1}{3}\gamma^{ab}\gamma^{cd}) + W^{abcd} \quad (3.38)$$

at such a point. The further specialization to the fully isotropic case is obtained if the invariant $W_{abcd}W^{abcd}$ vanishes in which case the last term drops out. Many crystals have cubical symmetry in their lowest energy (i.e. ground) state, although of course this symmetry will not be preserved when they are distorted.

4. THE STRAIN TENSOR AND THE HOOKEAN APPROXIMATION

Hookean elasticity theory is based on the supposition that there exists a locally relaxed state of the material, which we shall refer to as the *unstrained state*, in the sense that at each particular element $X \in \mathcal{X}$ there is a particular value κ_{ab} of the projection tensor γ_{ab} for which the energy per particle m takes a minimum value m_0 . The material tensor field κ_{ab} so defined, can be thought of as a canonical Riemannian metric on $\mathcal{X}$ which we shall refer to as the *unstrained reference tensor*. When interpreted as an orthogonal space-time tensor, κ_{ab} must satisfy the orthogonality condition

$$\kappa_{ab}u^b = 0 \quad (4.1)$$

and the material transport equation

$$[\kappa_{ab}]^* = 0. \quad (4.2)$$

There will be a corresponding inverse reference tensor field κ^{-1ab} on $\mathcal{X}$ to which there will be a corresponding orthogonal space-time tensor field satisfying

$$\kappa^{-1ab}u_b = 0, \quad (4.3)$$

$$[\kappa^{-1ab}]^* = 0, \quad (4.4)$$

$$\kappa^{-1ab}\kappa_{bc} = \gamma^a_c. \quad (4.5)$$

In addition to the material scalar field m_0 on $\mathcal{X}$ the unstrained state determines corresponding unstrained values ρ_0 of ρ and n_0 of n . When interpreted as space-time fields these three fields satisfy

$$\dot{m}_0 = 0, \quad (4.6)$$

$$\dot{n}_0 = 0, \quad (4.7)$$

$$\dot{\rho}_0 = 0. \quad (4.8)$$

The most convenient measure of the state of strain of the material, i.e. of its deviation from the unstrained state, is the *Lagrangian strain tensor* e_{ab} which is a symmetric orthogonal space-time tensor defined as half the difference between the actual value of γ_{ab} and the corresponding unstrained value κ_{ab} , i.e.

$$e_{ab} = \frac{1}{2}(\gamma_{ab} - \kappa_{ab}). \quad (4.9)$$

Clearly from the results of § 2 its material derivative is given by

$$[e_{ab}]^* = \theta_{ab} \quad (4.10)$$

and its material differential satisfies

$$d(e_{ab}) = \frac{1}{2} d(\gamma_{ab}). \quad (4.11)$$

It is reasonable to suppose that for small strains the equations of state can be represented by expansions in powers of the strain tensor components. The Hookean approximation consists of neglecting all but the lowest non-trivial powers of the strain tensor components. We shall refer to any model equation of state which has been simplified as far as possible by elimination of all but the lowest non-trivial powers of the strain tensor, as a *Hookean idealization*. Such an idealization can be used in place of the real equation of state of the medium, whenever the Hookean approximation is being applied.

The simplest way to obtain a Hookean idealization, is to specify the equation of state for the energy per particle as a quadratic function of the natural, i.e. covariant, components of the strain tensor, i.e.

$$m = m_0 + \frac{1}{2} K^{abcd} e_{ab} e_{cd}, \quad (4.12)$$

where K^{abcd} is a material tensor field which can clearly be chosen, without loss of generality, to have the symmetries

$$K^{abcd} = K^{(ab)(cd)} = K^{cdab}. \quad (4.13)$$

Since m_0 is the minimum value of m there are no linear terms. For the same reason K^{abcd} must be positive definite when thought of as a symmetric 6 by 6 matrix acting on the six independent components of e_{ab} .

When interpreted as an orthogonal space-time tensor, K^{abcd} satisfies the orthogonality condition

$$K^{abcd} u_d = 0 \quad (4.14)$$

and the material transport equation

$$[K^{abcd}]^* = 0. \quad (4.15)$$

It is easy to verify by an application of the formula (3.15) to the definition (4.12) that in the present case the structure tensor is given simply by

$$E^{abcd} = n K^{abcd}, \quad (4.16)$$

so that we may write

$$\rho = \frac{n}{n_0} \rho_0 + \frac{1}{2} E^{abcd} e_{ab} e_{cd}. \quad (4.17)$$

By application of (3.11), we obtain for the pressure tensor the very simple form

$$p^{ab} = -E^{abcd} e_{cd}. \quad (4.18)$$

It is instructive to compare the present formulation with that of Rayner (1963). In the formulation above the elasticity tensor satisfies the equation of motion

$$[E^{abcd}]^* = -\theta E^{abcd}. \quad (4.19)$$

In the Hookean theory of Rayner, a tensor c^{abcd} is introduced in effect by an equation formally identical with (4.18), i.e. by $p^{ab} = -c^{abcd}e_{cd}$ with p^{ab} and e_{ab} defined as in the present formulation. In an attempt to satisfy the conditions for perfect elasticity Rayner requires that his tensor c^{abcd} should satisfy the same symmetry conditions $c^{abcd} = c^{(ab)(cd)} = c^{cdab}$ as the tensor E^{abcd} defined here, but instead of the equation (4.19), Rayner requires that his tensor c^{abcd} should satisfy conditions equivalent to $[c_{ab}{}^{cd}]^* = 0$ which differs from (4.19), both in the use of the mixed rather than the contravariant form, and in the omission of the expansion term. If one works out the value of the elasticity tensor as defined by (3.12) using Rayner's formulation, one obtains

$$E^{abcd} = c^{abcd} + 2(\gamma^{a(c}c^{d)bef} + \gamma^{b(c}c^{d)aef})e_{ef} + \gamma^{cd}c^{abef}e_{ef}$$

which clearly does *not* satisfy the symmetry condition (3.16) owing to the presence of the final term. If Rayner had chosen to impose $[c_{ab}{}^{cd}]^* = \theta c_{ab}{}^{cd}$ this final term would have dropped out and the integrability condition (3.16) for perfect elasticity would have been satisfied. Thus the inconsistency in Rayner's formulation arises not from his use of the mixed rather than the contravariant form of c^{abcd} in the equation of motion (which merely leads to a more complicated expression for the function of state ρ) but in his omission of the expansion term (which has the effect of preventing ρ from being definable at all as a function of state). We should emphasize, however, that despite this weakness Rayner's formulation is perfectly adequate for the linearized small strain limit for which it was intended, since Rayner's equation of motion agrees with our equation (4.19) when the strain is zero (i.e. when the motion is rigid).

A special case of the present theory occurs when the medium is isotropic. In this case we can express the tensor K^{abcd} in the form

$$n_0 K^{abcd} = (\beta_0 - \frac{2}{3}\mu_0)\kappa^{-1ab}\kappa^{-1cd} + 2\mu_0\kappa^{-1a(c}\kappa^{-1d)b}, \quad (4.20)$$

$$\text{with} \quad \beta_0 = 0, \quad (4.21)$$

$$\dot{\mu}_0 = 0, \quad (4.22)$$

where the material constants β_0 and μ_0 represent the zero-strain *bulk modulus* and *modulus of rigidity* respectively.

5. THE SHEAR TENSOR AND THE HIGH-PRESSURE QUASI-HOOKEAN APPROXIMATION

The Hookean elasticity theory described in the previous section is inappropriate for the description of solid matter at high pressures—such as occur in the interiors of neutron stars—not only because of departures from linearity when the matter is far from its relaxed state, but also for the more serious reason that a fully relaxed state may not even exist within the limits of elastic behaviour. The crystalline structure in a neutron star crust, for example, owes its existence to the high pressure

of its environment. If one were to try to relax such a crystal its structure would break down (thereby introducing bulk viscosity into the equations of motion and so violating the conditions required for perfectly elastic behaviour), long before a state of zero stress could be attained. Thus under such conditions the fundamental material tensor κ_{ab} on which Hookean theory is based cannot even be defined. We must therefore cope not merely with situations in which the equations of state involve highly nonlinear dependence on the Lagrangian strain tensor but also with situations in which the Lagrangian strain tensor e_{ab} cannot be defined at all.

In order to deal with such situations the most natural way to proceed seems to be by considering states in which the material has not an absolute minimum of energy per particle m_0 , but a minimum of m subject to the restriction of constant particle number density n . We shall refer to such a state as an unsheared state. Thus at each material element $X \in \mathcal{X}$ we shall be led to introduce a family of positive definite tensor fields $\eta_{ab}(n)$ parametrized by n representing the value of the projection tensor γ_{ab} for the corresponding unsheared state with density n , in which the energy per particle m , and hence also the energy density ρ take minimum values for that particular value of n . We shall use the notation $\check{\rho}(n)$ for this minimum (i.e. unsheared) value of the energy density ρ so that the corresponding minimum value of the energy per particle m will be given by $\check{m}(n)$ where

$$n\check{m} = \check{\rho}. \quad (5.0)$$

When m has an absolute minimum m_0 so that the material tensor κ_{ab} and the scalars ρ_0 , and n_0 of the previous chapter are defined, we shall have

$$\eta_{ab}(n_0) = \kappa_{ab}, \quad (5.1)$$

$$\check{\rho}(n_0) = \rho_0, \quad (5.2)$$

$$\check{m}(n_0) = m_0. \quad (5.3)$$

We remark that in the most general circumstances there is no guarantee even that the unsheared states, and the corresponding material functions $\check{m}(n)$ and $\eta_{ab}(n)$ (which replace the unstrained state and the material constants m_0 and κ_{ab} of the Hookean case) will be well defined. One can imagine situations in which a crystal might owe its structure to the presence of a large, highly anisotropic stress, so that the reversible constant volume relaxation process which we have envisaged could not be carried out without the crystal structure breaking down. However, since exact calculations of neutron star structure have until now been based on the approximation that the stress is exactly isotropic (the perfect fluid idealization) it should be possible to make substantial improvements by using a theory capable of dealing with moderate deviations from isotropy, without worrying about extreme cases. In any case the primary objective of nearly all detailed studies of neutron star matter so far has been the estimation of the ground state energy density at given baryon number density, i.e. the estimation of $\check{\rho}(n)$. It is entirely in accord with the spirit of these studies to use a theory of elasticity based on the concept of continuous reversible deformation from the ground state. The only part of a neutron star for

which the present theory seems likely to be inadequate is the extreme surface of the crust where, according to Ruderman (1971 *a, b*), highly anisotropic molecular-type structures owing their existence to the presence of a strong magnetic field may be expected to be present. (Such a layer would fall outside the scope of the present theory independently of the question of whether the matter could be reversibly relaxed to a state of minimum energy at constant volume, due to the strong coupling with the electromagnetic field which could be expected to prevent the conservation equation (3.1) from being satisfied by the energy momentum tensor of the matter alone). Being very thin, such a layer would probably play a very minor role in the dynamics of the star.

Once $\check{m}(n)$ and $\eta_{ab}(n)$ have been defined as functions of state on $\mathcal{X}$, the scalar field n on $\mathcal{M}$ immediately determines a corresponding fixed scalar field $\check{m}$ and a corresponding fixed orthogonal space-time tensor field η_{ab} on $\mathcal{M}$. In addition to the orthogonality relation

$$\eta_{ab}u^b = 0 \quad (5.4)$$

they will satisfy equations of motion of the form

$$[\eta_{ab}]' = [\eta_{ab}]' \check{n}, \quad (5.5)$$

$$\check{m}' = \check{m}' \check{n}, \quad (5.6)$$

where we use primes to denote derivatives of functions of n with respect to n (which is itself of course a function of γ_{ab}). Only six of the seven components of $\check{\rho}$, $[\eta_{ab}]'$ can be specified independently by the equation of state since it follows from the definition of η_{ab} in terms of changes of state at constant volume, that its determinant is equal to the determinant of γ_{ab} , i.e.

$$\eta_c^c \{3\eta_a^b \eta_b^a - (\eta_a^a)^2\} = 2(3 - \eta_a^b \eta_b^c \eta_c^a), \quad (5.7)$$

so that its derivative with respect to n must satisfy

$$\eta^{-1ab}[\eta_{ab}]' = -2/n. \quad (5.8)$$

Here η^{-1ab} is defined as the inverse metric tensor to η_{ab} on $\mathcal{X}$. Interpreted as an orthogonal space-time tensor it satisfies

$$\eta^{-1ab}u_b = 0, \quad (5.9)$$

$$\eta^{-1ab}\eta_{bc} = \gamma^a_c. \quad (5.10)$$

It follows from (5.8) that we can introduce a new symmetric material tensor function of n , τ_{ab} defined by

$$n[\eta_{ab}]' = -\frac{2}{3}\eta_{ab} + \tau_{ab}, \quad (5.11)$$

which will be trace-free with respect to η_{ab} , i.e.

$$\eta^{-1ab}\tau_{ab} = 0. \quad (5.12)$$

Regarded as an orthogonal space-time tensor field it will of course satisfy

$$\tau_{ab}u^b = 0. \quad (5.13)$$

We shall refer to τ_{ab} as the *compressional distortion tensor*. It determines the way in which the crystal deformation deviates from conformal contraction when it undergoes compression by varying isotropic pressure. Since it will always determine at least one uniquely preferred direction, it is evident that this compressional distortion tensor must vanish not only for isotropic material but also for any material whose crystal structure has cubical symmetry. The ionic crystal structure (cf. Dyson 1971) to be expected in neutron star crusts can almost certainly be expected to have cubical symmetry, and therefore in the practical applications we have in mind it will probably be safe to set τ_{ab} equal to zero.

We can express the equations of motion for m and η_{ab} or η^{-1ab} by

$$\dot{\mathcal{M}} = -n\theta\dot{\mathcal{M}}', \quad (5.14)$$

$$[\eta_{ab}]' = \theta(\frac{2}{3}\eta_{ab} - \tau_{ab}), \quad (5.15)$$

$$[\eta^{-1ab}]' = -\theta(\frac{2}{3}\eta^{-1ab} - \eta^{-1ac}\eta^{-1bd}\tau_{cd}), \quad (5.16)$$

in terms of the six independent components of $\mathcal{M}'$ and τ_{ab} .

In order to measure deviations from the state of minimum energy at fixed particle number density we now introduce the *constant volume shear tensor* s_{ab} defined as half the difference between the actual value of γ_{ab} and the corresponding value η_{ab} which it would have for the unsheared state at the same volume, i.e.

$$s_{ab} = \frac{1}{2}(\gamma_{ab} - \eta_{ab}). \quad (5.17)$$

This tensor has only five independent components, since in consequence of (5.7) it satisfies the identity

$$s_a{}^a = -\frac{s_a{}^b s_b{}^a - \frac{4}{3}s_a{}^b s_b{}^c s_c{}^a}{1 - s_a{}^a - 2s_a{}^b s_b{}^a + \frac{2}{3}(s_a{}^a)^2}. \quad (5.18)$$

Thus the trace $s_a{}^a$ of s_{ab} and hence also its contraction $\eta^{-1ab}s_{ab}$ must vanish to first order (although not exactly) when expressed in terms of expansions in powers of s_{ab} . This constant volume shear tensor obeys the equation of motion

$$[s_{ab}]' = \sigma_{ab} + \theta(\frac{2}{3}s_{ab} + \frac{1}{2}\tau_{ab}), \quad (5.19)$$

where σ_{ab} is the rate of shear tensor defined by (2.20). Its material differential satisfies

$$d[s_{ab}] = \frac{1}{2}d[\gamma_{ab}] + (\frac{1}{3}\eta_{ab} - \frac{1}{2}\tau_{ab})\frac{dn}{n}. \quad (5.20)$$

Just as in Hookean theory we expanded the energy per particle m about its absolute minimum value, so here we shall expand m about its minimum value $\mathcal{M}(n)$ at fixed number density n in powers of s_{ab} . The high pressure analogue of the Hookean approximation is the approximation in which all powers of s_{ab} but the lowest are ignored. When this approximation is applicable it will be sufficient to use in place of the real equation of state, an idealized equation of state based on an equation for m which is quadratic in the shear. Thus we define a *quasi-Hookean idealized equation of state* by

$$m = \mathcal{M} + \frac{1}{2}B^{abcd}s_{ab}s_{cd}, \quad (5.21)$$

where $B^{abcd}(n)$ is a material tensor function of state which, like $\check{m}(n)$ is determined as a function of n only. We can suppose without loss of generality that $B^{abcd}(n)$ satisfies the same symmetry conditions

$$B^{abcd} = B^{(ab)(cd)} = B^{cdab} \quad (5.22)$$

as the structure tensor. We shall also require that $B^{abcd}(n)$ is trace free on contraction with the corresponding tensor $\eta_{ab}(n)$ on either the first or last pair of indices, i.e.

$$B^{abcd}\eta_{cd} = 0. \quad (5.23)$$

This condition does not involve any effective loss of generality since as we have remarked s_{ab} is trace free to first order under contraction with η^{-1ab} due to the fact that both η_{ab} and γ_{ab} have the same determinant so that omission of the traces of B^{abcd} which might otherwise be included in (5.21) only affects the value of m to cubic and higher order in s_{ab} . If we did not impose (5.23), B^{abcd} would contain redundant information at quadratic order in s_{ab} . With this restriction B^{abcd} has 15 independent components as compared with the 21 independent components of the corresponding Hookean tensor K^{abcd} .

When interpreted as an orthogonal space-time tensor, B^{abcd} satisfies the orthogonality relation

$$B^{abcd}u_d = 0, \quad (5.24)$$

and the equation of motion

$$[B^{abcd}]' = -n\theta[B^{abcd}]'. \quad (5.25)$$

Just as the derivatives of η_{ab} are restricted by (5.11), so also, in consequence of (5.23), the derivatives of B^{abcd} are restricted by

$$n[B^{abcd}]' \eta_{cd} = -B^{abcd}\tau_{cd}. \quad (5.26)$$

Thus the idealized high-pressure equation of state (5.21) is specified by 21 independent functions of n , namely the independent components of η_{ab} and B^{abcd} and $\check{m}$.

The general expression for the elastic structure tensor E^{abcd} corresponding to the equation of state (5.21) is rather complicated, but its unsheared value $\check{E}^{abcd}(n)$ (i.e. its value when s_{ab} is zero) can be expressed fairly simply in terms of the corresponding unsheared values $\check{S}^{abcd}(n)$, $\check{D}^{ab}(n)$ and $\check{E}(n)$ of the quantities S^{abcd} , D^{ab} , E introduced in the decomposition (3.19). Thus from (3.12) we obtain

$$\check{S}^{abcd}(n) = nB^{abcd} + 2\check{p}(\eta^{-1c(a}\eta^{-1b)d} - \frac{1}{3}\eta^{-1ab}\eta^{-1cd}), \quad (5.27)$$

$$\check{D}^{ab}(n) = \frac{1}{2}nB^{abcd}\tau_{cd}, \quad (5.28)$$

$$\check{E}(n) = 9\beta - 3\check{p} + \frac{9}{4}nB^{abcd}\tau_{ab}\tau_{cd}, \quad (5.29)$$

where we have used the notation

$$\check{p}(n) = n^2\check{m}' = n\check{p}' - \check{p} \quad (5.30)$$

for the unsheared value $\check{p}(n)$ of the pressure p at density n , and

$$\beta(n) = n\check{p}' = n^2\check{p}'' \quad (5.31)$$

for the *bulk modulus* at density n .

Hence using (3.7) and (3.9) we can write the expressions for the energy density and pressure tensor of matter obeying the idealized quasi-Hookean equation of state (5.21) in the form

$$\rho = \check{\rho} + \frac{1}{2}nB^{abcd}s_{ab}s_{cd}, \quad (5.32)$$

$$p^{ab} = \check{p}\gamma^{ab} - (nB^{abcd} + \gamma^{ab}\check{D}^{cd})s_{cd} + \frac{1}{2}n^2\gamma^{ab}[B^{efcd}]'s_{ef}s_{cd} \quad (5.33)$$

The expressions (5.27), (5.28), (5.29) are useful for showing how the high pressure quasi-Hookean equation of state (5.21) corresponds to the Hookean equation of state (4.12) in the low strain limit if the latter exists. Thus using a suffix zero to denote the value of a function of n evaluated at n_0 (when this latter is defined) and noting that

$$\check{p}_0 = 0, \quad (5.34)$$

we see that the tensor K^{abcd} introduced in (4.12) must be given by

$$n_0 K^{abcd} = n_0 B_0^{abcd} + (\kappa^{-1ab}\check{D}_0^{cd} + \kappa^{-1cd}\check{D}_0^{ab}) + (\beta_0 + \frac{1}{2}\check{D}_0^{ef}\tau_{ef})\kappa^{-1ab}\kappa^{-1cd} \quad (5.35)$$

in order that (4.12) should agree with (5.21) in the low strain limit. Conversely, given K^{abcd} we can obtain the appropriate value of $\check{D}_0^{ab}$ from

$$\check{D}_0^{ab} = \frac{1}{3}n_0(K^{abcd}\kappa_{cd} - \frac{1}{3}K^{cdef}\kappa_{cd}\kappa_{ef}\kappa^{-1ab}), \quad (5.36)$$

and hence obtain B_0^{abcd} from

$$n_0 B_0^{abcd} = n_0 K^{abcd} - (\kappa^{-1ab}\check{D}_0^{cd} + \kappa^{-1cd}\check{D}_0^{ab}) - \frac{1}{9}n_0 K^{efgh}\kappa_{ef}\kappa_{gh}\kappa^{-1ab}\kappa^{-1cd}. \quad (5.37)$$

If required, the relevant value of τ_{0ab} can then be calculated by finding the solution (which will be unique except in degenerate cases) of

$$B_0^{abcd}\tau_{cd} = (2/n_0)\check{D}_0^{ab} \quad (5.38)$$

and hence finally the zero-strain bulk modulus can be obtained from the formula

$$\beta_0 = \frac{1}{9}n_0 K^{abcd}\kappa_{ab}\kappa_{cd} - \frac{1}{2}\check{D}_0^{ab}\tau_{0ab}. \quad (5.39)$$

A useful measure of the departure of a quasi-Hookean elastic solid from adiabatic perfect fluid behaviour is the invariant $\mu(n)$ which we shall refer to as *modulus of rigidity* at density n defined by

$$\mu^2(n) = \frac{1}{20}n^2 B^{abcd} B^{efgh} \eta_{ae} \eta_{bf} \eta_{cg} \eta_{dh}. \quad (5.40)$$

It is clear that $\mu(n)$ is zero if and only if the tensor B^{abcd} is zero, i.e. if and only if the material offers no resistance to deformation at constant density. From (5.40) we find that the zero strain modulus of rigidity, at zero pressure, μ_0 is given by

$$20\mu_0^2 = n_0^2 K^{abcd} K^{efgh} (\kappa_{ae}\kappa_{bf} - \frac{2}{3}\kappa_{ab}\kappa_{ef})\kappa_{cd}\kappa_{gh} + \frac{1}{9}n_0^2 (K^{abcd}\kappa_{ab}\kappa_{cd})^2. \quad (5.41)$$

6. THE PERFECT SOLID

In any solid material whose crystal structure possesses cubical symmetry with respect to its unstrained states, the compressional distortion tensor τ_{ab} must, as has already been remarked, be zero. By applying this condition to (5.15) and (5.16) we see that in such a material the reference tensors η_{ab} and η^{-1ab} satisfy the simple equations of motion

$$[\eta_{ab}]' = \frac{2}{3}\eta_{ab}\theta, \quad (6.1)$$

$$[\eta^{-1ab}]' = -\frac{2}{3}\eta^{-1ab}\theta, \quad (6.2)$$

and by (5.19) the shear-strain tensor satisfies

$$[s_{ab}]' = \frac{2}{3}s_{ab}\theta + \sigma_{ab}. \quad (6.3)$$

We remark that (6.1) and (6.2) can be integrated explicitly to give

$$\eta_{ab} = (n/n_0)^{-\frac{2}{3}}\kappa_{ab}, \quad (6.4)$$

$$\eta^{-1ab} = (n/n_0)^{\frac{2}{3}}\kappa^{-1ab}, \quad (6.5)$$

where κ_{ab} , κ^{-1ab} and n_0 are material constants, which may be identified with the unstrained reference tensor and its inverse and the corresponding unstrained number density, as defined in § 4, provided they exist.

In this section we wish to consider the particular case of the intrinsic elastic properties of a solid material which is not merely cubically symmetric but actually *isotropic* with respect to its unstrained states. Such a situation can be expected to arise in any solid whose fine-scale crystal structure is highly irregular (as is that of ordinary glass for example). It will also hold, at least as a useful approximation, in many cases where there is a highly regular cubically symmetric lattice. In any such case the energy per particle m can be a function only of the particle number density n and the two independent scalar invariants of the shear tensor. It is convenient to choose these two invariants to be the minimal positive definite quadratic invariant s^2 defined by

$$s^2 = \eta^{-1ab}\eta^{-1cd}s_{bc}s_{da} - \frac{1}{3}(\eta^{-1ab}s_{ab})^2 \quad (6.6)$$

and the cubic invariant

$$w = \eta^{-1ab}\eta^{-1cd}\eta^{-1ef}s_{bc}s_{de}s_{fa}. \quad (6.7)$$

All invariant power series functions of the shear tensor can be expressed in terms of power series functions of s^2 and w since the shear has no linear invariant (the quantity $\eta^{-1ab}s_{ab}$ being equal to $s^2 + \frac{4}{3}w$ + terms of higher order). We prefer to work with contractions in terms of the reference tensor η^{-1ab} rather than with the analogous invariants

$$s_a{}^b s_b{}^a - \frac{1}{3}(s_a{}^a)^2 \quad \text{and} \quad s_a{}^b s_b{}^c s_c{}^a$$

defined in terms of contractions with γ^{ab} because the latter have more complicated properties under differentiation with respect to γ_{ab} than s^2 and w as defined above.

Thus we can express the conditions for a perfect solid to have locally isotropic intrinsic properties by the requirement that (6.1) (and hence also equations (6.2) to (6.5)) be satisfied, and that along each particle world line the mass per particle is determined by an equation of state of the form

$$m = F(n, s^2, w), \quad (6.8)$$

where $F(n, s^2, w)$ (which can of course vary from one particle to another) is a well behaved function of the three variables n, s^2, w such that the quantities

$$\check{\rho}(n) = nF(n, 0, 0), \quad (6.9)$$

$$\beta(n) = n^3 \partial^2 F / \partial n^2 (n, 0, 0) + 2n^2 \partial F / \partial n (n, 0, 0), \quad (6.10)$$

$$\mu(n) = n \partial F / \partial (s^2) (n, 0, 0) \quad (6.11)$$

are all positive. We shall refer to any material satisfying these conditions as a *perfect solid*, the quantities $\rho(n)$, $\beta(n)$ and $\mu(n)$ being its *unsheared energy density*, *bulk modulus*, and *modulus of rigidity* respectively.

The general formula for the density ρ and pressure tensor p^{ab} in the perfect solid will take the form

$$\rho = nF(n, s^2, w) \quad (6.12)$$

$$\begin{aligned} p^{ab} = & \left\{ n^2 \frac{\partial F}{\partial n} (n, s^2, w) + n s^2 \left(\frac{2}{3} \frac{\partial F}{\partial (s^2)} (n, s^2, w) + \frac{\partial F}{\partial w} (n, s^2, w) \right) \right. \\ & \left. - n(2w - \frac{1}{3}(\eta^{-1cd} s_{cd})^2) \frac{\partial F}{\partial w} (n, s^2, w) \right\} \gamma^{ab} - 2n \frac{\partial F}{\partial (s^2)} (n, s^2, w) (\eta^{-1a(c} \eta^{-1b) d} \\ & - \frac{1}{3} \eta^{-1ab} \eta^{-1cd}) s_{cd} - 3n \frac{\partial F}{\partial w} (n, s^2, w) \eta^{-1a(c} \eta^{-1b) d} \eta^{-1ef} s_{ce} s_{df}. \end{aligned} \quad (6.13)$$

Introducing the notation

$$\check{p} = n^2 (\partial F / \partial n) (n, 0, 0) \quad (6.14)$$

for the zero-shear strain value of the pressure we can express the corresponding value $\check{E}^{abcd}(n)$ of the elasticity tensor E^{abcd} in the state of zero-shear strain, by

$$\check{E}^{abcd} = (\beta - \frac{1}{3}\check{p}) \eta^{-1ab} \eta^{-1cd} + 2(\mu + \check{p}) (\eta^{-1a(c} \eta^{-1d) b} - \frac{1}{3} \eta^{-1ab} \eta^{-1cd}). \quad (6.15)$$

The general formula (6.8) specializes to a *quasi-Hookean perfect solid* equation of state if it is linearized in s^2 and if the w dependence is left out altogether, i.e. if we set

$$F(n, s^2, w) = \check{m} + \mu s^2 / n, \quad (6.16)$$

where $\check{m}(n)$ and $\mu(n)$ are the functions of number density only, defined above. This is equivalent to setting

$$B^{abcd} = 2(\mu/n) (\eta^{-1a(c} \eta^{-1d) b} - \frac{1}{3} \eta^{-1ab} \eta^{-1cd}) \quad (6.17)$$

in the quasi-Hookean equation of state (5.21), from which it can be verified that the definition of μ introduced here by (6.11) is consistent with the definition given in the previous section by (5.40). By applying (6.12) and (6.13) we see that in the isotropic case the quasi-Hookean equations (5.32) and (5.33) reduce to the expressions

$$\rho = \check{\rho} + \mu s^2, \quad (6.18)$$

$$p^{ab} = \{\check{p} + (n\mu' - \frac{1}{3}\mu) s^2\} \gamma^{ab} - 2\mu \{\eta^{-1a(c} \eta^{-1d) b} - \frac{1}{3} \eta^{-1ab} \eta^{-1cd}\} s_{cd}, \quad (6.19)$$

for the energy momentum tensor components in a quasi-Hookean perfect solid.

APPENDIX. CONVECTED FRAMES AND CONVECTED DERIVATIVES

It is evident from the definition in § 2 that a differential one-form orthogonal to the flow lines in $\mathcal{M}$ is materially transported (i.e. has zero convective derivative) if and only if its components in terms of a co-moving coordinate system are constant along the world lines, i.e. (as can be seen by considering the particular case of a *properly* co-moving coordinate system) if and only if it is Lie transported by the flow field u^a . It follows that for an orthogonal differential form, and hence also for any covariant orthogonal space-time tensor field on $\mathcal{M}$, material transport is the same as Lie transport with respect to the flow. Thus for a covariant orthogonal space-time tensor field the *convected derivative is the same as the Lie derivative with respect to the flow field*, i.e.

$$\begin{aligned} [T_{a\dots}]' &= \mathfrak{L}_u(T_{a\dots}) \\ &= T_{a\dots;c} u^c + T_{c\dots} u^c{}_{;a} + \dots, \end{aligned} \quad (\text{A } 1)$$

where $T_{a\dots}$ is a covariant orthogonal tensor field on $\mathcal{M}$ and $\mathfrak{L}_u$ denotes the Lie derivative with respect to u^a .

In order to obtain the corresponding formula for contravariant and mixed tensors we make use of the fact that the indices of orthogonal space-time tensors can be raised by contraction with the inverse projection tensor field γ^{ab} .

The mixed form of this tensor corresponds under $\mathcal{P}$ to the Kronecker tensor on $\mathcal{X}$ and therefore it clearly must have zero convected derivative in $\mathcal{M}$, i.e.

$$[\gamma_a{}^b]' = 0. \quad (\text{A } 2)$$

Also, by direct application of the formula (A 1) we obtain

$$[\gamma_{ab}] = 2u_{(a;b)} + 2\dot{u}_{(a} u_{b)}. \quad (\text{A } 3)$$

Hence, by differentiating (2.9) using the Liebnitz rule, we obtain

$$[\gamma^{ab}]' = -2u^{(a;b)} - 2\dot{u}^{(a} u^{b)}. \quad (\text{A } 4)$$

Thus, again using Liebnitz rule, we obtain

$$\begin{aligned} [T_{a\dots}{}^{b\dots}]' &= [T_{a\dots} \gamma^{b\dots}]' \\ &= T_{a\dots;c}{}^{b\dots} u^c + T_{c\dots}{}^{b\dots} u^c{}_{;a} + \dots - T_{a\dots}{}^{c\dots} (u^b{}_{;c} + u^b \dot{u}_c) - \dots, \end{aligned} \quad (\text{A } 5)$$

where $T_{a\dots}{}^{b\dots}$ is a mixed orthogonal tensor field on $\mathcal{M}$. This is the formula of Oldroyd (1970). It can be re-expressed in the form

$$[T_{a\dots}{}^{b\dots}]' = \mathfrak{L}_u(T_{a\dots}{}^{b\dots}) - T_{a\dots}{}^{c\dots} \dot{u}_c u^b - \dots. \quad (\text{A } 7)$$

In order to see how this formula should be generalized to apply to general (i.e. not necessarily orthogonal) tensor fields on $\mathcal{M}$ we introduce a convected 4-frame field on $\mathcal{M}$. As explained in § 2, a convected 4-frame field on $\mathcal{M}$ is defined in terms of a material 3-frame field on $\mathcal{X}$. Let us choose such a field, locally to consist of three

vectors e^a , $i = 1, 2, 3$, and three covectors e_a , $j = 1, 2, 3$ related by the standard duality conditions

$$e^a e_a = \delta_{(i)}^{(j)} \quad (\text{A } 8)$$

where $\delta_{(i)}^{(j)}$ is a Kronecker symbol. The corresponding orthogonal vector and covector fields in $\mathcal{M}$ will be material constants, and hence by (A 5) will satisfy the equations of motion

$$e^a_{(i);c} u^c - e^c u^a_{(i);c} - e^c \dot{u}_c u^a = 0, \quad (\text{A } 9)$$

$$e_a^{(i)};_c u^c + e_c u^a_{(i);a} = 0, \quad (\text{A } 10)$$

as well as the orthogonality conditions

$$e^a u_a = 0, \quad (\text{A } 11)$$

$$e_a u^a = 0, \quad (\text{A } 12)$$

where i runs over 1, 2, 3. In order to extend the 3-frame to a 4-frame we must choose a fourth vector e^a and a corresponding covector e_a on $\mathcal{M}$ in such a way that the duality conditions (A 8) are supplemented by standard Lorentz signature duality conditions of the form

$$e_a e^a = -\delta_{(0)}^{(i)}, \quad (\text{A } 13)$$

$$e^a e_a = -\delta_{(i)}^{(0)}, \quad (\text{A } 14)$$

where $\delta_{(0)}^{(i)}$, $\delta_{(i)}^{(0)}$ are Kronecker symbols, and now i runs over the values 0, 1, 2, 3. It is clear from (A 11) and (A 12) that these requirements allow us no freedom (except for a choice of sign) in specifying e^a and e_a . We therefore set

$$e_a = u_a, \quad (\text{A } 15)$$

$$e^a = u^a. \quad (\text{A } 16)$$

By differentiating the orthonormality condition (2.7) we obtain

$$u_c u^c_{;a} = 0, \quad (\text{A } 17)$$

$$\dot{u}_c u^c = 0. \quad (\text{A } 18)$$

It is easy to check, using (A 11) and (A 12) that these equations of motion for u_a and u^a can be combined with the equations (A 9) and (A 10) in the form

$$\dot{e}^a = e^c (u^a_{(i);c} + u^a \dot{u}_c), \quad (\text{A } 19)$$

$$\dot{e}_a^{(i)} = -e_c (u^c_{(i);a} + u^c \dot{u}_a), \quad (\text{A } 20)$$

where i now runs over the full range 0, 1, 2, 3. We can use these equations of motion for the convected 4-frame to evaluate the convected derivative of a general space-time tensor according to the definition

$$[T_{a\dots}^{b\dots}]_{(i)}^{(j)} e^a e_b \dots = (T_{a\dots}^{b\dots} e^a e_b \dots)_{(i)}^{(j)}, \quad (21)$$

whereby the 4-frame components of the convected derivative are the proper time derivatives of the 4-frame components. Working out the right-hand side we obtain

$$\begin{aligned} (T_{a\dots}^{b\dots} e^a e_b \dots)_{(i)}^{(j)} &= \dot{T}_{a\dots}^{b\dots} e^a e_b + T_{a\dots}^{b\dots} e^c (u^a_{;c} + u^a \dot{u}_c) e_b \dots + \dots \\ &\quad - T_{a\dots}^{b\dots} e^a e_c (u^c_{;b} + u^c \dot{u}_b) \dots - \dots, \end{aligned} \quad (\text{A } 22)$$

which leads directly to the general formula (2.20) given in § 2.

The authors would like to thank G. F. R. Ellis for helpful advice and discussions.

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